

Archaeologist

16+3 · WEIGHT 2 · LONGEST (Lukas Moroz)

CLEAR priors → inventum (L_ctx + deep SS + multimodal + VisionMix) → lineage
I-JEPA→...→2.1→2-AC. Label every claim CLEAR vs TENTATIVE.

Spine: Unlock dense features in JEPA WM line · L_ctx + deep SS + VisionMix/ViT-G ·
I-JEPA→V-JEPA→2→2.1→2-AC · Grasp 60→80% · nav ~10× · triangle SigLIP/DINO|MAE|JEPA

FAIR Meta · Unlock dense features in the JEPA world-model line · [facebookresearch/vjepa2](https://facebookresearch.com/vjepa2)

Talk arc — densest lineage pack (Lukas)

Priors → inventum → triangle → robotics deepen → labels

01-03

Spine + CLEAR priors

AMI · I-JEPA · V-JEPA · 2/2-AC · BYOL · DINO · Registers · NWM · DROID

04-07

Inventum + paper figs

mask-only diagnosis · L_ctx · deep SS · Fig 1/3/4

08-11

Lineage + triangle + recipe

I-JEPA → ... → 2.1 → 2-AC · SigLIP|MAE|JEPA · Fig 5

12-14

Robotics deepen + nav

state estimation not new planner · Fig 8/9 · +20pp / 10×

15-16

CLEAR/TENTATIVE + close

π_0 /OpenVLA/GR00T = rhyme · hybrid = proposal · tattoo

Shared spine — tattoo every deck

Six claims that must land in every seat

1 Unlock dense features in the JEPA world-model line — L_ctx + deep SS + VisionMix/ViT-G.

2 CLEAR lineage: I-JEPA → V-JEPA → V-JEPA 2 → V-JEPA 2.1 → 2-AC.

3 Franka grasp 60→80%; latent nav ~10× cheaper than NWM.

4 Triangle: SigLIP/DINO | MAE | JEPA WM (now dense).

5 Must: NYU 0.307; ADE 22.2→47.9; Ego4D 7.71; Tables 6-7; L_ctx alone hurts globals / deep SS restores.

6 SSv2 77.7; EK100 40.8; Tartan ATE 5.687 / 10.6s vs NWM 103.2s.

CLEAR priors — cited / direct parents

Extend I-JEPA pack; do not contradict CLEAR/TENTATIVE labels

LeCun AMI 2022	Names JEPA; predictive embeddings + AMI agenda	CLEAR
I-JEPA Assran 2023	Image JEPA; multi-block; emb L2; EMA — pack sibling	CLEAR
V-JEPA Bardes 2024	Video feature-prediction JEPA (2404.08471)	CLEAR
V-JEPA 2 / 2-AC 2025	Scale + AC planning on DROID; Franka CEM recipe	CLEAR
BYOL / EMA stop-grad	Collapse avoidance reused in y-encoder	CLEAR
DINOv2 / DINOv3	Dense-feature foil & table baselines (cited)	CLEAR
Registers (Darcet)	Context-as-register hypothesis for mask-only JEPA	CLEAR
LVD-142M / NWM / DROID	VisionMix images · nav baseline · AC robot data	CLEAR
data2vec / MAE / VideoMAE	Mask-predict / generative foils (related work)	CLEAR

This inventum — CLEAR (what 2.1 adds)

Diagnosis → dense loss → deep SS → scale → robotics payoff · KANTA reconcile

Diagnosis

Mask-only L_predict → context tokens act as registers → weak dense (ADE 22.2 / NYU 0.682).

L_ctx dense loss

Supervise ALL tokens; distance-weighted L1 near masks. ADE 22.2→33.8; NYU 0.682→0.474.

Deep self-sup

4-level predictor restores globals (SSv2 62.5→72.1) while lifting dense further.

Multimodal tok

Native 2D/3D patchify + modality emb — kills image-as-static-video hack.

VisionMix163M

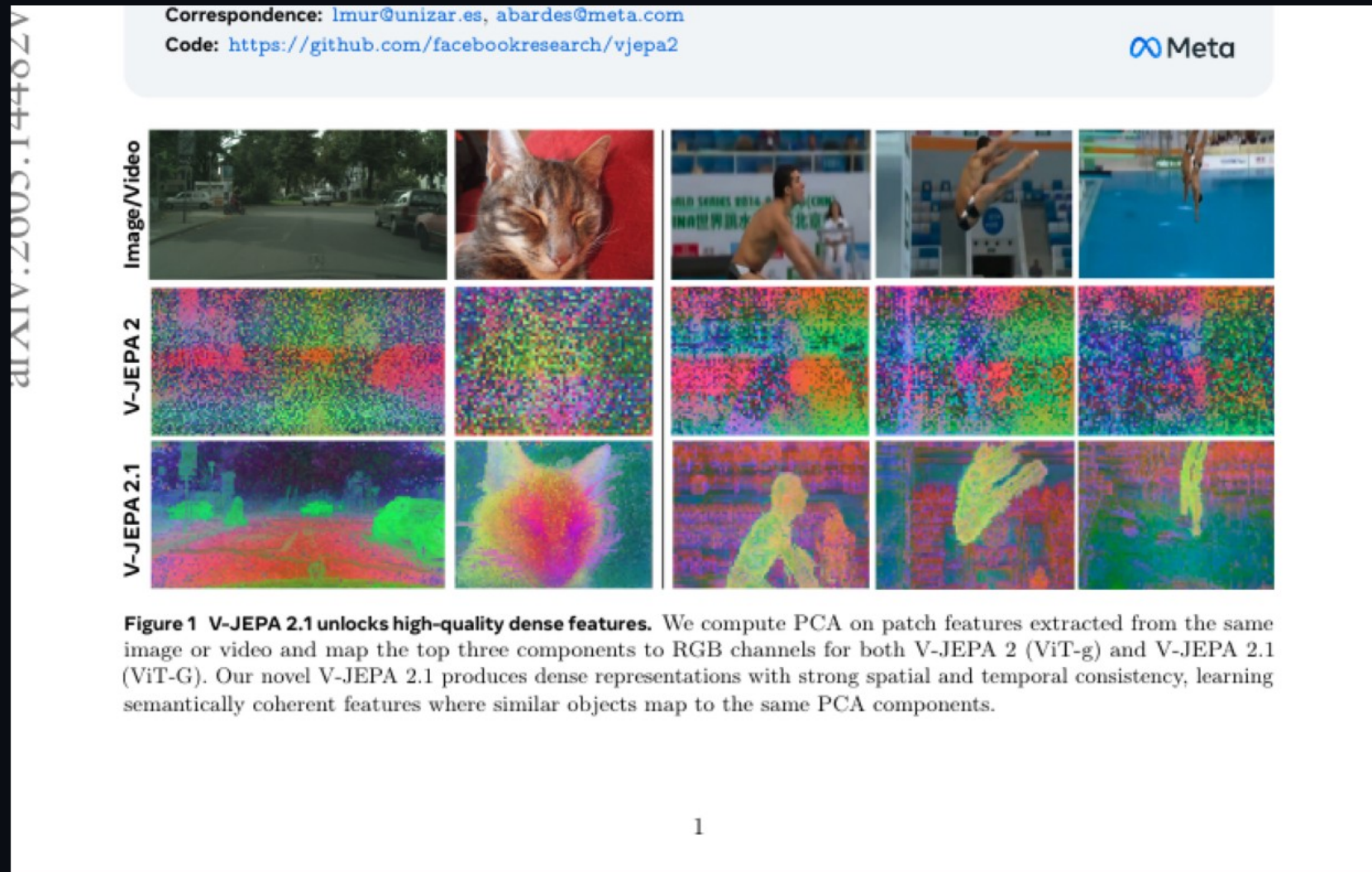
LVD-142M in; ImageNet out; YT/SSv2 reweight → ViT-G + cool-down.

Payoff

Grasp 60→70→80 (n=10); ~10× latent nav; distill B/L for deploy.

Fig. 1 — dense unlock visual (CLEAR)

PCA RGB: V-JEPA 2 noisy → V-JEPA 2.1 coherent parts + temporal consistency

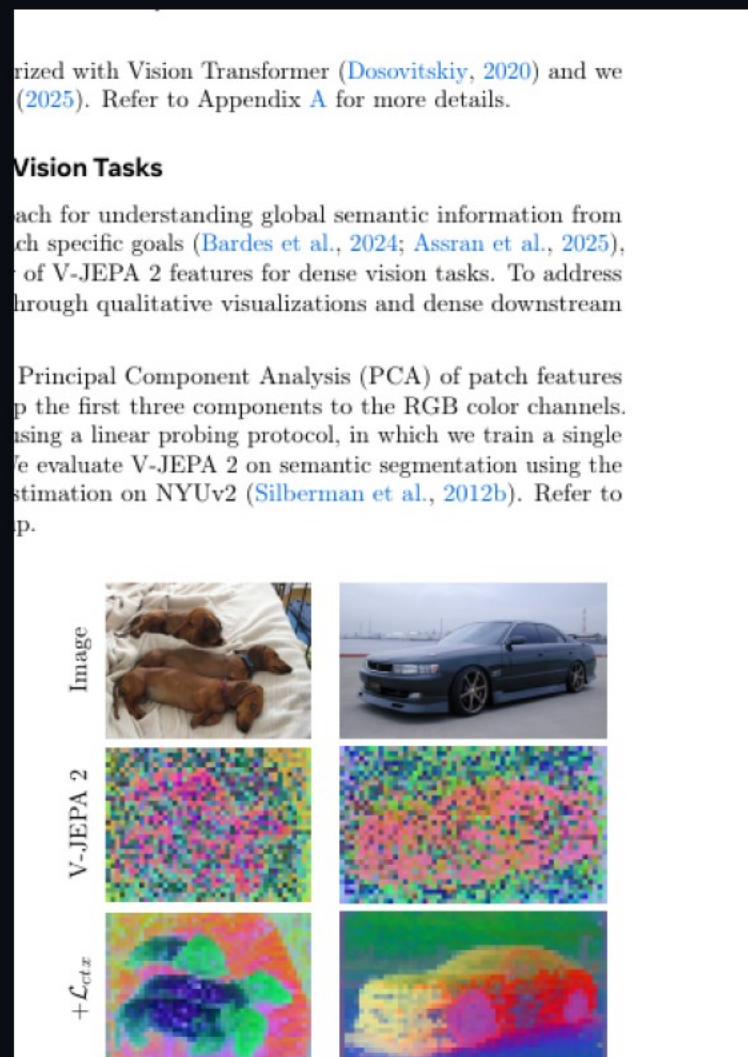


1

Paper Fig. 1 — title claim made visual. Similar objects map to same PCA components.

Fig. 3 — causal evidence for L_ctx (CLEAR)

Dogs/car PCA: V-JEPA 2 static → +L_ctx coherent semantic parts



Paper Fig. 3 — context-as-register hypothesis → supervise visible tokens near masks.

Fig. 4 — architecture (CLEAR inventum)

Multimodal tok · multi-level x-enc · $L_{\text{dense}} = L_{\text{predict}} + L_{\text{ctx}} \cdot \text{EMA } y\text{-enc}$

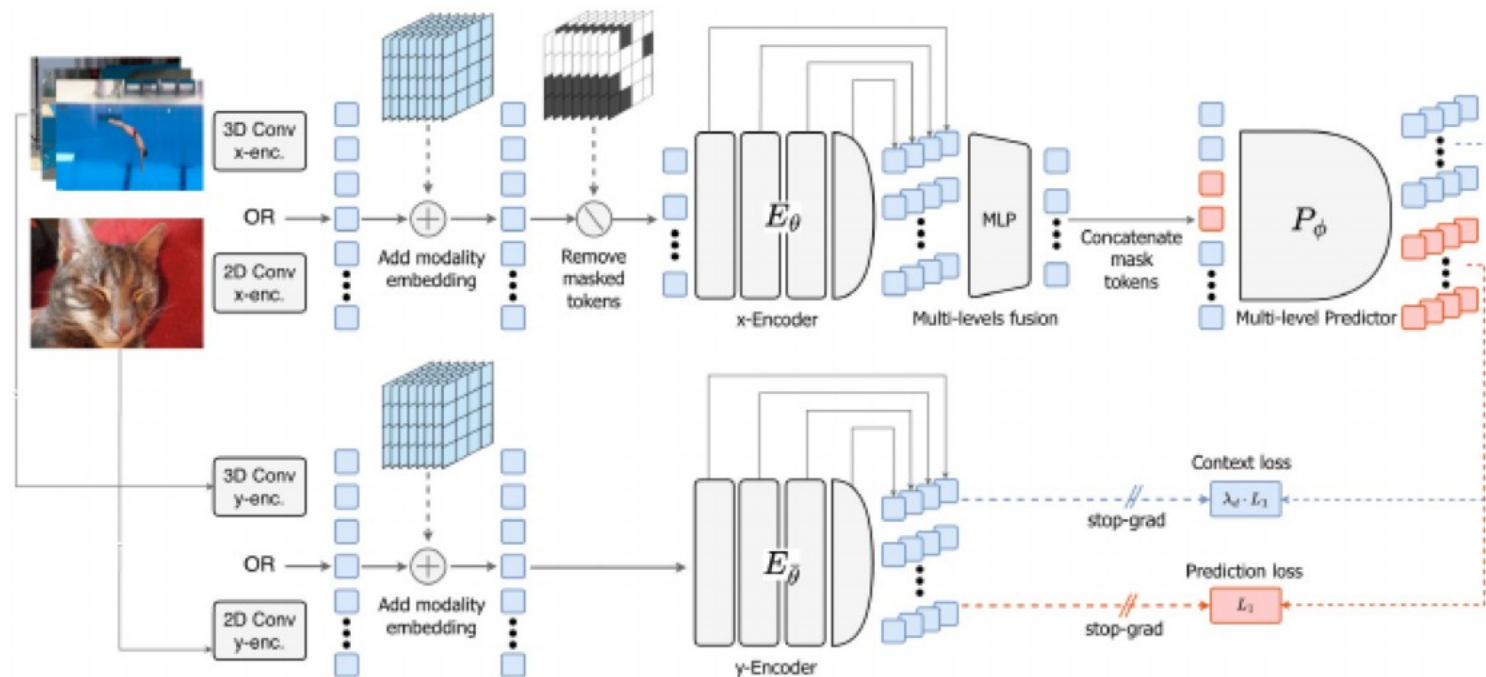


Figure 4 V-JEPA 2.1 Detailed Architecture. Images and videos are processed by respectively either a 2D or 3D Convolutional patch embedding. Then, 3D Rotational Positional Encoding (RoPE) and learnable modality embedding are added. The x -encoder processes the visible tokens and outputs multi-level embeddings formed by concatenating the normalized output from intermediate encoder blocks. Then, a MLP fuses this multi-level representation and reduces its dimensionality. These context tokens are concatenated with learnable mask tokens carrying spatio-temporal positional information. The predictor processes the combined sequence and produces multi-level predictions for the masked tokens. Training uses two different losses: (i) an L1 loss on masked-token predictions (the original V-JEPA objective), and (ii) a distance-weighted L1 loss on nearby context tokens, both supervised using the y -encoder multi-level outputs.

Paper Fig. 4 — 2D/3D tokenizer + 3D RoPE; predictor emits 4-level preds; weighted context L1.

Lineage chain — CLEAR session mnemonic

I-JEPA → V-JEPA → 2 → ★2.1★ → 2-AC



Moral (extend I-JEPA pack)

I-JEPA proved latent multi-block prediction. V-JEPA moved that to time. V-JEPA 2 showed planning. V-JEPA 2.1's punchline: masked-only JEPA under-supervises visible tokens → weak dense maps; supervise everything (weighted) + deep SS restores the dense/global Pareto. 2-AC recipe reused with denser encoder.

Consistency triangle — JEPA side now dense

Unchanged structure from I-JEPA pack; 2.1 fills the dense cell

DINOv2/v3 / SigLIP

Discriminative dense
(image-first JEA)

→ VLA vision towers
(π_0 , OpenVLA, ...)

MAE / VideoMAE

Generative MIM
pixels / tokens

→ strong FT
weaker linear probes

I-JEPA→...→2.1→2-AC

Predictive latents
+ NOW dense

→ planning WM
Franka / latent nav

Moral: VLAs still ship SigLIP/DINO (left). JEPA's robotics payoff is world-model / planning (right).
 π_0 /OpenVLA/GR00T = design rhyme NOT citation unless paper cites them (it cites DINO as dense foils only).

Fig. 5 / Table 1 — recipe stack (CLEAR)

L_ctx lifts ADE but tanks SSv2; deep SS restores globals; scale+cool-down finish

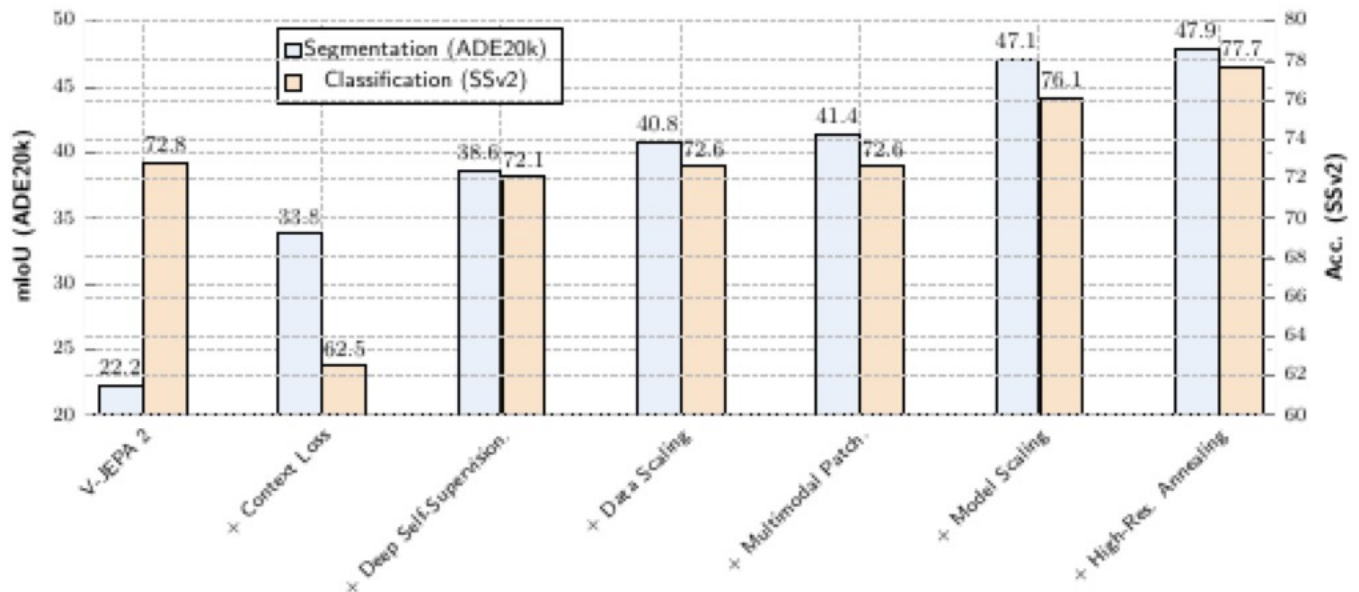


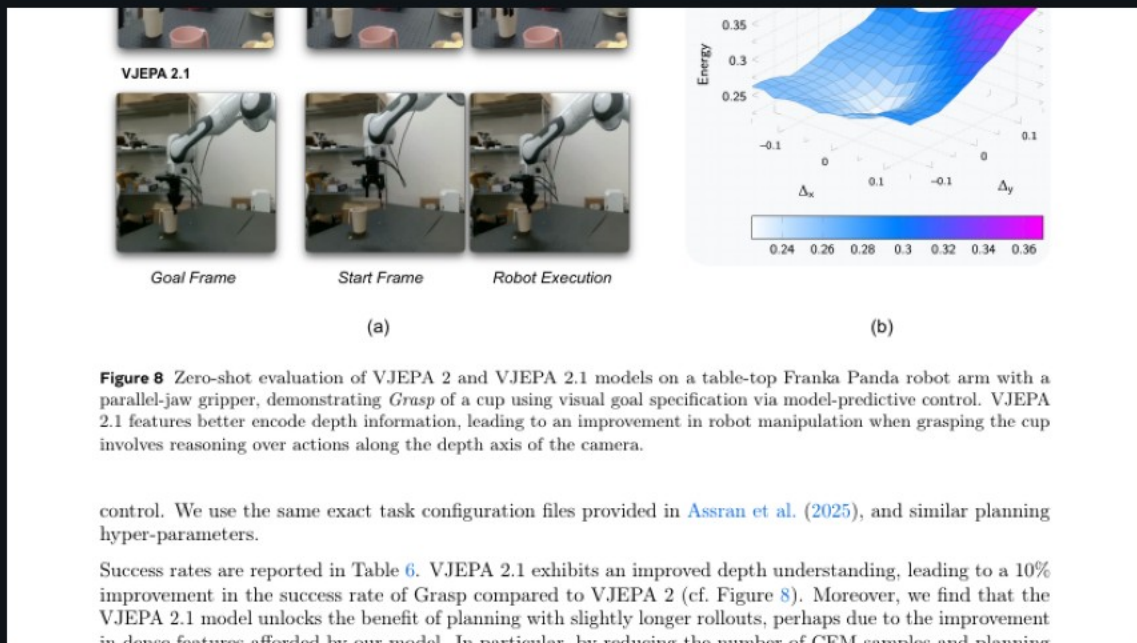
Figure 5 Impact of individual components of our novel V-JEPA 2.1 training recipe. The ablation is conducted starting from a ViT-L architecture, on single-image semantic segmentation on ADE20k, and action classification on SSv2. Introducing *Weighted-Context Self-Supervision* with the context loss significantly improves segmentation, at the cost of classification. *Deep Self-Supervision* allows to retrieve the performance, and further improving segmentation. Scaling the image data with VisionMix 163M dataset, combined with a *Multi-Modal Tokenizer* further improve the results. Finally, the recipe *scales* with model size and high resolution *cool-down*.

V-JEPA 2.1 (ViT-L) · mIoU 22.2 · Acc. 72.8 · VisionMix 163M · Multi-Modal Tokenizer · V-JEPA 2.1 (ViT-L) · mIoU 47.9 · Acc. 77.7

ADE 22.2→47.9 · SSv2 72.8→77.7 · MUST: L_ctx alone hurts globals / deep SS restores.

Robotics deepen — state estimation, not new planner

CLEAR: denser encoder makes longer CEM useful (Fig. 8 / Table 6)



CLEAR inventum for robots

Not a new planner — same AC codebase + CEM as V-JEPA 2. Payoff = depth/local structure so horizon-8 rollouts help (grasp 60→70 matched → 80% H=8).

Failures shift to gripper timing, not spatial misunderstanding.

n=10 cup tasks — promising, narrow. Cite Tables 6–7 only.

I-JEPA pack extend: durable idea = predictor-as-WM; first CLEAR Meta demo = 2-AC. This pack: dense features were the missing piece for precise spatial/depth under the same planner.

Latent nav — ~10× cheaper than NWM (CLEAR Table 7)

CDiT on 2.1 reps · 8 DDIM vs ≥ 128 SD-VAE · Tartan ATE 5.687 / 10.6s

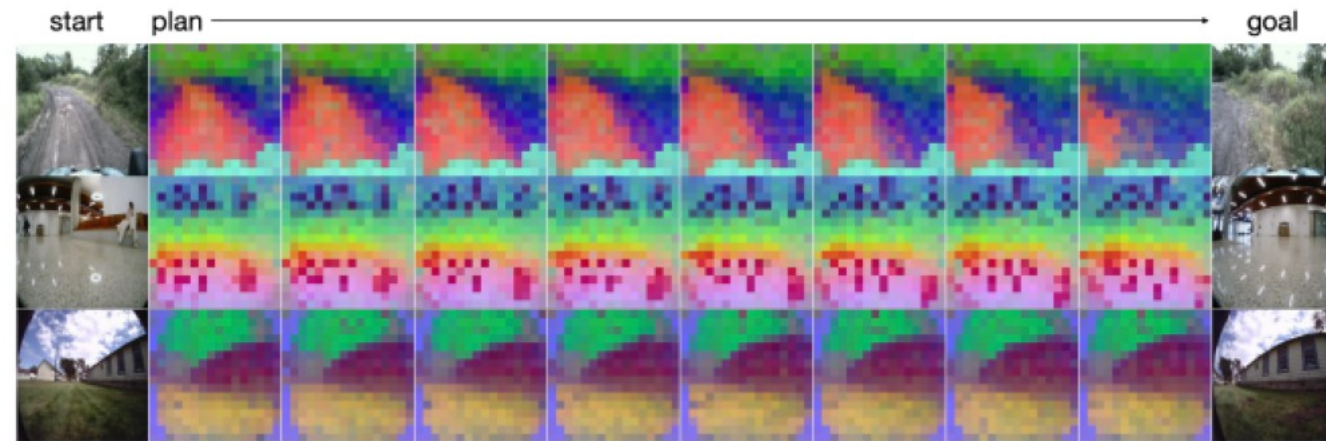


Figure 9 Planning Navigation Trajectories in Latent Space. V-JEPA 2.1 enables 10× faster and more accurate navigation planning compared to (Bar et al., 2025). We show PCA visualizations of 8 denoising steps of the planned latent trajectory between a start frame and a goal frame.

Table 7 Open Loop Navigation Planning. We finetune V-JEPA 2.1 on robot navigation datasets and compare the open-loop planning performance to NWM. Reporting the average planning time (seconds), normalized Average Trajectory Error (ATE) and Relative Trajectory Error (RTE). V-JEPA 2.1 representation enables 10× speed-up without sacrificing performance.

Fig. 9 — PCA planned trajectories. System claim (repr + sampler); still CLEAR in-paper number.

CLEAR vs TENTATIVE — label every claim

Exam discipline · extend I-JEPA pack · do not contradict

CLEAR — say it

- AMI → I-JEPA → V-JEPA → 2 → 2.1 → 2-AC
- L_ctx + deep SS inventum + Tables 1-2
- NYU 0.307 · ADE 47.9 · Ego4D 7.71 · SSv2 77.7
- Franka T6 60→80%; Tartan T7 5.687 / 10.6s
- Registers hypothesis cited; DINO as foil
- facebookresearch/vjepa2 code+weights

TENTATIVE / FORBIDDEN

- π_0 /OpenVLA/GR00T 'are JEPA children'
- Hybrid JEPA+FM as demonstrated result
- Franka n=10 = general manip SOTA
- ADE = dense SOTA (trails DINOv3 55.9)
- Drop-in SigLIP replacement without probes
- BEHAVIOR q-score ↑ from citing 0.307

Numbers tattoo — pin these

Reconciled with KANTA_PACK · shared across seats

0.307

NYUv2 RMSE linear
beats DINOv3-7B 0.309

22.2→47.9

ADE path T1
33.8→38.6→47.9

7.71

Ego4D STA mAP
ViT-G

40.8

EK100 Action R@5
ViT-G

77.7%

SSv2 attentive
72.8→62.5→72.1→77.7

60/70/80

Franka grasp n=10
matched→H=8 CEM

5.687

Tartan Drive ATE
10.6s vs NWM 103.2s

~10×

latent nav cheaper
than NWM SD-VAE

L_ctx

alone hurts globals
deep SS restores

33.8/62.5

T2 weighted $\lambda=0.5$
ADE / SSv2 trade-off

ViT-G 2B

VisionMix163M
+ cool-down 64@384

distill B/L

ViT-L $\approx$ G on
many probes (T11)

Archaeologist — close

Takeaways + exam bite

1. CLEAR inventum: L_ctx + deep SS + multimodal tok + VisionMix — unlocks dense JEPA features.
2. Lineage CLEAR: I-JEPA → V-JEPA → 2 → 2.1 → 2-AC; robotics = state-est upgrade, not new planner.
3. Triangle: SigLIP/DINO | MAE | JEPA-now-dense — VLAs left; WM/planning right.
4. TENTATIVE: π_0 /OpenVLA/GR00T rhyme only; hybrid JEPA+FM = proposal; n=10 Franka narrow.

EXAM BITE

V-JEPA 2.1 = dense-state upgrade to Meta's video WM line; cite Tables 6–7 for robots; hybrid JEPA+FM not demonstrated here.